

Eigenvalues and inverse problems

Rank-two maximizers of spectral spread on an entry interval

Difficulty: challenging **Importance:** interesting to the community**Status:** Partially resolved

Rating rationale: Challenging reflects an all-dimensions rank structure for global spread maximizers; community impact is an extremal spectral problem for entry-bounded symmetric matrices.

Reviewed continuation — 13 September 2026

Sidney Holden (Center for Computational Biology, Flatiron Institute, Simons Foundation) supplies a proof for the unrestricted strip $-1 \leq a \leq -1 + (1 + \sqrt{2})/n^2$ in every dimension, with a rank-two endpoint maximizer. The [continuation report](#), Theorems 1–2 and the boundary-strip proof, also bounds all endpoint matrices whose centered sign matrix has rank at most two. Its nonnegative-parameter case includes exact rational polynomial-positivity certificates.

Status remains **Partially resolved**: outside the strip, arbitrary centered sign matrices of rank at least three remain untreated. Rank at most two for the centered sign matrix is not a hypothesis available for every maximizer.

The [independent informal Codex AI-agent review](#) records its accepted scope and checks. [Submission, original package, reproduction and verified affiliation](#). This is not external human peer review or formal verification; no Lean verification was performed.

Further partial results — 12 September 2026

Sidney Holden (Center for Computational Biology, Flatiron Institute, Simons Foundation) supplies a [signed-threshold reduction and exact certificates](#) proving maximum spreads $\sqrt{73}$, $\sqrt{133}$, and $\sqrt{161}$ for $(n, a) = (8, 1/2), (10, 0), (11, 0)$, respectively. Explicit rank-two endpoint matrices attain each bound. These finite cases supplement the previously cited ranges; they do not prove the conjecture for all n and a .

The partial results passed an [independent informal Codex AI-agent audit](#), including exact certificate and coverage checks. Status remains Partially resolved; the remaining dimensions and interval parameters are unresolved. [Submission, certificate reproduction and verified affiliation](#). No novelty, external human peer review or formal verification is claimed.

Problem statement

For an integer $n \geq 2$ and $a \in [-1, 1]$, let

$$\mathcal{S}_n[a, 1] = \{A \in \mathbb{R}^{n \times n} : A = A^T, a \leq A_{ij} \leq 1 \text{ for all } i, j\}.$$

The spread of A is $s(A) = \lambda_{\max}(A) - \lambda_{\min}(A)$. The Fallat–Xing conjecture asserts that there exists a matrix B of rank exactly two, every entry of which belongs to $\{a, 1\}$, such that

$$s(B) = \max_{A \in \mathcal{S}_n[a, 1]} s(A).$$

All diagonal entries are free to vary within the same interval as the off-diagonal entries. The assertion concerns existence of a rank-two maximizer; it does not assert that every maximizer has rank two. The normalization $[a, 1]$, $-1 \leq a < 1$, is the source’s reduction of the arbitrary nondegenerate real-interval problem. All dimensions and intervals form one problem.

Why it matters

The conjecture reduces an eigenvalue optimization over a full symmetric matrix family to a low-rank extremizer. It sharpens spectral bounds obtainable from entrywise uncertainty alone.

References

S. M. Fallat and Y. Xing, *On the spread of certain normal matrices*, Linear and Multilinear Algebra 60 (2012), 1391–1407, §2. N. J. Calkin, R. M. Corless, L. Gonzalez-Vega, J. R. Sendra, and J. Sendra, *On the maximal spread of symmetric Bohemian matrices*, 2025, §1 (displayed Fallat–Xing conjecture), §§3.2, 7–8, and §10.

Earlier status check — 2026-09-08

Version 1 of the 2025 paper is the latest listed version checked. It proves the conjecture for all $a \in (-1, 1)$ when $2 \leq n \leq 7$, and for $a = 0$ when $2 \leq n \leq 8$ or 3 divides n . The case $a = -1$ was already proved by Zhan. Its conclusion explicitly records a gap in the authors’ attempted general proof. The [authors’ supporting repository](#) confirms the finite computational ranges. Searches for *Fallat Xing spread conjecture proof 2026* and later maximal-spread papers found no general resolution. Restricting entries to interval endpoints alone is already known and is not counted separately.

Audit update — 2026-09-10

Rechecked [Corless and collaborators](#), especially the proved cases and conclusion. The conjecture holds for all displayed interval parameters through order seven, with additional $a = 0$ dimensions, but the general all-orders statement remains open. Spread/rank-two searches found no subsequent completion. The status now explicitly records these substantive proved ranges rather than describing the entire entry simply as open.